

LAMINATED OBJECT MANUFACTURING



COURSE CONTENT

- 2.2.1 Introduction**
- 2.2.2 Laminated Object Manufacturing (LOM)**
- 2.2.3 LOM Materials**
- 2.2.4 Applications**

Prof. Shiva Krishna J

Dept. of Mechanical & Automation Engg., Agnel Institute of Technology and Design, Assagao

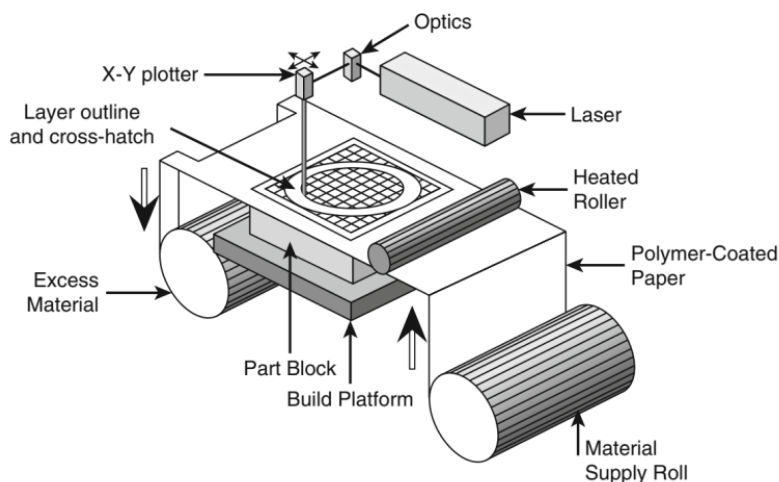
2.2.1 INTRODUCTION

In Section 1.3: Solid-Based Processes, Lamination-Based Processes were introduced. In this section, the Laminated Object Manufacturing (LOM) process is discussed in detail.

Several other processes have been developed based on the principle of sheet lamination, utilizing different build materials and cutting strategies. Due to the nature of their construction, only the outer contours of the parts are typically cut. The process sequence may vary sheets can either be cut first and then stacked (or) stacked first and then cut.

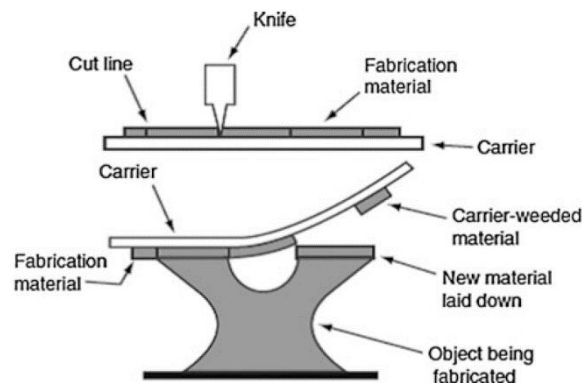
Among the various sheet lamination techniques, one of the most popular methods employs paper as the build material, bonded using a polymer-based adhesive. The typical paper thickness used ranges from 0.07 mm to 0.2 mm. Sheet Lamination process is also referred as Layer Laminate Manufacturing. The sheet lamination processes are classified as

1. **Bond-Then-Form Processes:** Sheet material layers are bonded together before cutting them into the desired shape to create a 3D geometry.



Bond-Then-Form Processes

2. **Form-Then-Bond Processes:** sheet material is cut to shape first and then bonded to the substrate. This approach is popular for construction of parts in metallic or ceramic materials that are thermally bonded.



Form-Then-Bond Processes

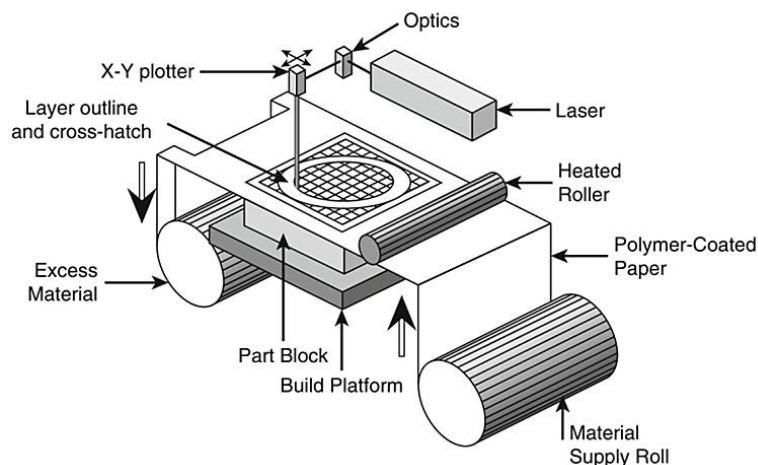
2.2.2 LAMINATED OBJECT MANUFACTURING (LOM)

LOM is a solid-based additive manufacturing process originally developed by Helisys Inc. (now Cubic Technologies). The LOM process fabricates 3D models by laminating layers

of adhesive-coated sheet material typically paper (or) plastic foils and cutting each layer to shape using a laser. The process produces parts with good surface finish and rigidity, suitable for concept modeling, tooling, and pattern making.

HARDWARE

- Build Platform: The platform serves as the base on which the part is fabricated layer by layer. After each layer is completed, the platform moves downward in the Z-direction to allow deposition and lamination of the next sheet.
- Material Supply and Take-up Rolls: The build material is provided in the form of continuous sheets wound on a supply roll. Fresh material is drawn from this roll, laminated onto the build platform, and after cutting, the take-up roll collects the waste or excess sheet material.
- Laminating Roller: A heated roller applies pressure and heat to bond the newly fed sheet to the previously laminated layer. This ensures strong adhesion and minimizes delamination during cutting.
- Laser Cutting System: A CO₂ laser is used to cut the cross-sectional outline of the layer according to CAD data, as well as to create crosshatch patterns in surrounding waste areas for easy removal after fabrication.



Hardware and Principle of LOM

- The process begins with the CAD model of the part, which is converted into an STL file. The software slices the model into thin layers and generates tool paths for cutting each contour.
- A fresh sheet of adhesive-coated material is fed onto the build platform. The heated roller bonds it to the previous layer. The laser then cuts the outline of the part's cross-section and hatches the excess material into a grid pattern.
- After each layer is cut, the platform lowers by one sheet thickness, and the next sheet is advanced, laminated, and cut. This process repeats until all layers are stacked and bonded.
- The final output is a solid block containing the fully enclosed part. The surrounding crosshatched waste material is manually removed to extract the model. The resulting wood-like LOM part can be finished using traditional post-processing methods such as sanding, polishing, painting, or sealing.

ADVANTAGES:

1. **Wide Range of Materials:** LOM can utilize a diverse range of organic and inorganic sheet materials, including paper and plastic films, making it a versatile process for various prototyping needs.
2. **High Build Speed:** Since the laser only traces the outer contour of each layer rather than scanning the entire surface, LOM offers faster build times, especially for parts with large, solid cross-sections.
3. **Good Dimensional Accuracy:** LOM systems typically achieve a dimensional accuracy better than ± 0.127 mm, providing sufficient precision for conceptual prototypes.
4. **No Need for Separate Support Structures:** The surrounding uncut material naturally supports the part during fabrication, eliminating the need for additional support structures.
5. **Cost-Effective for Large Models:** The use of inexpensive materials such as paper and relatively fast build rates makes LOM a cost-effective solution for producing large, non-functional prototypes and casting patterns.

LIMITATIONS:

1. **Unsuitable for Thin-Walled Features:** The process is not ideal for producing delicate or thin-walled geometries, particularly along the Z-axis.
2. **Limited Mechanical Strength:** Since layers are bonded primarily through adhesive and thermal lamination, the interlayer strength is relatively low. As a result, LOM parts are generally not suitable for load-bearing applications.
3. **Labor-Intensive Post-Processing:** After fabrication, the part must be manually separated from the surrounding crosshatched waste material, often using tools similar to wood-carving instruments.
4. **Surface Finish and Anisotropy:** The final part often exhibits a layered wood-grain appearance and may require additional finishing operations such as sanding or sealing.

2.2.3 LOM MATERIALS

LOM utilizes sheet-based feedstock materials that are adhesive-coated and bonded layer-by-layer through heat and pressure. The choice of material depends on the intended application, desired mechanical strength, surface finish, and post-processing requirements.

Commonly used Materials are:

1. Paper Sheets: The most used material in LOM is paper sheets, which produce parts with a wood-like appearance and texture. These sheets typically have a thickness ranging from 0.07 mm to 0.2 mm and are bonded using a polymer-based adhesive. Paper-based LOM parts are particularly suitable for concept models and visual prototypes, offering good surface finish and dimensional stability at low cost.
2. Plastic Films: In addition to paper, plastic films such as Polyvinyl Chloride (PVC), Polycarbonate (PC), and Polypropylene (PP) are also used in LOM systems. These materials provide greater durability, improved moisture resistance, and smoother surfaces compared to paper-based builds, making them suitable for applications that require enhanced strength and surface quality.

2.2.4 APPLICATIONS

LOM is primarily used for conceptual modeling, aesthetic visualization, and tooling applications, where mechanical performance is less critical than geometry and appearance.

- Conceptual and Design Prototypes: Used for visual models, form and fit testing, and design verification in early stages of product development. Offers good surface finish and dimensional accuracy for large-scale parts.
- Casting and Tooling Patterns: Ideal for producing patterns for sand casting, investment casting, and vacuum forming molds. Paper or composite LOM parts can be infiltrated or coated to improve strength and surface hardness.
- Architectural and Display Models: Used to create architectural mock-ups, scale models, and exhibition prototypes due to low cost and fast fabrication of large geometries.
- Educational and Research Models: Useful in academic and training environments to demonstrate additive manufacturing principles and to create geometric prototypes.